Wachirawit Piyaprapapan

Data & AI Engineer | ML Systems | Data Platforms | Applied AI

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Summary

AI engineering student and part-time AI engineer focused on production-oriented ML, RAG systems, data platforms, and edge computer vision. Experienced in building FastAPI/Next.js applications, PostgreSQL-backed pipelines, forecasting models, and observability for AI systems.

Education

Chulalongkorn University, B.Eng. in Electrical Engineering – Bangkok, Thailand — Aug 2022 – May 2026

- GPAX: 3.44 / 4.00; Second-class honours.
- Relevant coursework: Data Science, Data Engineering, Estimation, Statistical Learning, Optimization.
- Capstone: Generative Video-Based Sky Image Forecasting for Thai sky images using VQ-VAE-based video prediction.

Skills

Languages: Python, SQL, JavaScript, Bash

ML / AI: PyTorch, Scikit-learn, LightGBM, spaCy, OpenCV, Hugging Face, RAG, time-series forecasting, computer vision

Backend / Data: FastAPI, Next.js, React, PostgreSQL, SQLModel, Pandas, PySpark, Airflow, dbt

MLOps / Tools: Docker, Git, Grafana, Prometheus, GCP, Vercel, Selenium

Experience

AI Engineer, Part-time, Hobbit Technologies – Bangkok, Thailand — Feb 2026 – present

- Develop and maintain RAG services with LiteLLM and FastAPI, using Pydantic-validated structured outputs to reduce downstream parsing failures.
- Architect a modular Next.js/PostgreSQL platform for CEFR-aligned content generation and evaluation, including automated grammar-correction workflows.
- Improve LLM output consistency through prompt templates, schema validation, and REST API service boundaries.

AI Engineer Intern, Hobbit Technologies – Bangkok, Thailand — June 2025 – Aug 2025

- Engineered a Dockerized computer-vision data engine with OpenCV, automated drift checks, and active-learning triggers, reducing labeling cost by approximately 20k THB/year.
- Deployed Grafana/Prometheus monitoring for Edge AI inference on on-premise IoT sensors, improving visibility into production-line failures.
- Built data-cleaning and dataset-versioning workflows for edge computer-vision experiments.

Electrical Engineering Intern, AGC Flat Glass – Bangkok, Thailand — June 2024 – Aug 2024

- Built a data-driven control-logic prototype that integrated production data into PLC workflows.
- Analyzed machine and production signals to support operational-efficiency improvement of approximately 10%.

Projects

Generative Video-Based Sky Image Forecasting for Thai Sky Images

- Implemented and optimized a VQ-VAE-based forecasting model with Charbonnier loss, perceptual loss, EMA codebook updates, and sub-pixel interpolation decoding.
- Achieved 37.32 dB peak validation PSNR, a 16% improvement over baseline, while compressing channel-sequence representations for more stable temporal prediction.

Coffee Chain Demand Forecasting System

- Engineered horizon-safe lag, rolling, event-proximity, calendar, and store-category demand features for daily product-level sales forecasting across representative coffee stores.
- Reduced internal walk-forward CV MAE by 12.2% over the leakage-safe baseline and identified local-event and store-category behavior as the strongest business signals.

On-Demand Delivery Data Platform & Decision Intelligence System

- Built ETL/ELT workflows with Python, PostgreSQL, Airflow, dbt, and Docker to ingest raw JSON events into analytics-ready tables and feature-store structures.
- Added schema validation and time-aware model validation to support ETA and delay-prediction experiments.

Football Player Value Forecasting & Similarity Recommendation System

- Developed time-series regression and clustering pipelines to forecast player market value and group players by role similarity.
- Translated forecasting errors and cluster structures into recruitment and benchmarking insights.

Competitions

Microsoft AI Engineering Skills and Hackathon for Employment 2026 2026

- Reached finalist team stage among 2,000+ participants.
- Built a dashboard-driven emergency-case intake and evaluation pipeline using Azure Foundry and Azure OpenAI.

Thailand Super AI Engineer Season 6 — Coffee Chain Time-Series Hackathon 2026

- Led research and feature engineering for daily coffee-sales forecasting; designed seasonal and event-driven demand features.
- Contributed technical evidence for the pitching track by connecting forecasting features to inventory and order-optimization workflows.

Thailand Super AI Engineer Season 6 — Edge AI for Intelligent Transport Systems 2026

- Developed a computer-vision data pipeline for traffic monitoring and anomaly detection.
- Used VLM-assisted inspection and semi-supervised learning techniques to bootstrap custom traffic datasets.

I-squared Hackathon — Motorbike Rider Anomaly Detection & Classification

- Built a two-stage computer-vision pipeline using YOLOv8 for detection and ViT for classification.
- Designed automated data-cleaning and image-enhancement workflows; reached semifinals with over 90% training accuracy and over 70% accuracy on unseen real-world images.